

LUND UNIVERSITY

Mathematics, Faculty of Science

Number theory
MATM35
HT 2020
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Written examination (online)
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1. Show that 3 is a primitive root modulo 17. Compute a corresponding table of indices. Use this to solve the congruence

$$3x^6 \equiv 7 \pmod{17}.$$

2. Determine the infinite continued fraction representation of the number $\sqrt{35}$.

3. Prove that

$$\sum_{d|n} \sigma(d) = \sum_{d|n} \frac{n}{d} \tau(d)$$

for all n, where τ is the divisor function and σ is the sum-of-divisors function.

4. Find, or prove that there do not exist:
   (a) three consecutive positive integers $n, n+1, n+2$, none of which can be written as the sum of two squares;
   (b) two consecutive positive integers $n, n+1$, none of which can be written as the sum of three squares.

5. Find (with proof) the 4 smallest odd primes p for which the congruence

$$x^2 + 7 \equiv 0 \pmod{p}$$

has a solution.

6. Let p be a prime such that $p \equiv 3 \pmod{4}$. Prove that

$$\sum_{a=1}^{p-1} a^2 \left(\frac{a}{p}\right) = p \sum_{a=1}^{p-1} a \left(\frac{a}{p}\right).$$

Give careful motivations for your solutions!